GENERALISED FRESH-QUANTIFIED PERMUTATION FIXED POINTS DECOMPOSE INTO FRESHNESS AND ALGEBRAIC INVARIANCE

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Editorial note. Source https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.FSCD.2026.10. Copyright Ali K. Caires-Santos, Maribel Fernández, Murdoch James Gabbay and Daniele Nantes-Sobrinho, CC BY4.0. Theorem2.3.5 and complete author appendix proof/support faithfully transcribed. Original shorthand references and notation are retained without added assumptions or proofs.

Allowed standard prerequisites (editorial).

- Finite permutation groups, support equivariance and the single-atom Some/Any principle. Publisher sections2.1-2.3 and cited Proposition4.10 [16]. Actual nested Some/Any and cycle-support proofs are included, as is the whole two-part generalisation.

Permutations, support and fresh quantification (publisher sections 2.1–2.3)

We fix disjoint countably infinite sets: the set of atoms $\mathbb{A} = \{a, b, c, \ldots\}$ and unknowns $\mathbb{V} = \{X, Y, Z, \ldots\}$. A (finite) permutation on $\mathbb{A}$ is a bijection $\pi : \mathbb{A} \to \mathbb{A}$ whose set of moved atoms $\mathrm{mv}(\pi) = \{a \in \mathbb{A} : \pi(a) \neq a\}$ is finite. Write $\pi \circ \pi'$, id and π^{-1} for functional composition, the identity, and the inverse of permutations, respectively. The group of all finite permutations of atoms is denoted $G(\mathbb{A})$. Every finite permutation $\pi \in G(\mathbb{A})$ can be uniquely decomposed (up to reordering) into pairwise disjoint cycles as $\pi = \gamma_1 \circ \cdots \circ \gamma_n$, with $\mathrm{mv}(\gamma_i) \cap \mathrm{mv}(\gamma_j) = \varnothing$ for $i \neq j$. This is called the canonical form of π.

Assumption 2.1.1. Permutations are always in a canonical form. A nominal set is a pair $(\mathcal{X}, \cdot)$ consisting of a set $\mathcal{X}$ and an action $\cdot : G(\mathbb{A}) \times \mathcal{X} \to \mathcal{X}$ such that every element $x \in \mathcal{X}$ has finite support, i.e., there exists a finite $B \subseteq \mathbb{A}$ with $\pi \cdot x = x$ for all $\pi \in G(\mathbb{A})$ such that $\pi(a) = a$ for every $a \in B$. The least such B is denoted $\mathrm{supp}(x)$. An atom a is fresh for x (written $a\#x$) if $a \notin \mathrm{supp}(x)$. Sometimes, when there is no ambiguity, we refer to a nominal set $(\mathcal{X}, \cdot)$ by its carrier set $\mathcal{X}$ and leave the action $\cdot$ implicit. Throughout this section, let $\mathcal{X}$ denote a nominal set. We consider formulas of the form $\phi(c, \bar{x})$, where c is an atom and $\bar{x} = x_1, \ldots, x_n$ is a finite list of parameters with each $x_i \in \mathcal{X}$. Typically, ϕ is a predicate in higher-order logic [22] or ZFA set theory [14,16].

Definition 2.3.1. The N-quantifier is formally defined as:

$$\mathsf{N}\, c.\phi(c, \bar{x}) \iff \{c \in \mathbb{A} : \phi(c, \bar{x})\} \text{ is cofinite.}$$

A cofinite subset of a set M is a subset B whose complement in M is a finite set. In cases where multiple fresh atoms are introduced at once, we consider nested N-quantification of the form $\mathsf{N}\, c_1 \cdots \mathsf{N}\, c_n.\phi(c_1, \ldots, c_n, \bar{x})$, abbreviated as $\mathsf{N}\, \bar{c}.\phi(\bar{c}, \bar{x})$. For a finite set of atoms A and a finite list of elements $\bar{x} = x_1, \ldots, x_k$ of $\mathcal{X}$, write $A\#\bar{x}$ as abbreviation for $A \cap \mathrm{supp}(\bar{x}) = \varnothing$, where $\mathrm{supp}(\bar{x}) = \bigcup_{j=1}^{k} \mathrm{supp}(x_j)$. If

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$\bar{a} = a_1, \ldots, a_n$ is a list of atoms, we write $\bar{a}\#x$ as shorthand for $\{\bar{a}\}\#x$, where $\{\bar{a}\} = \{a_1, \ldots, a_n\}$. A fundamental connection between the N-quantifier and freshness is given by the following equivalence [16]:

$$a\#x \iff \mathsf{N}\, c.(a\ c)\cdot x = x.$$

We define a generalised fixed-point identity as an assertion of the form $\mathsf{N}\,\bar{c}.\pi\cdot x = x$, where π is a permutation that may contain atoms bound by N.
Definition 2.3.2 ($\bar{c}$-cycles and $\bar{c}$-permutations). Let π be a permutation written as a product of disjoint cycles and let $\bar{c}$ be a list of atoms.

(1) A cycle η of π is called a $\bar{c}$-cycle if $\mathrm{mv}(\eta)\cap\{\bar{c}\} \neq \varnothing$, and a $\neg\bar{c}$-cycle otherwise.
(2) A permutation π is a $\bar{c}$-permutation iff all its cycles are $\bar{c}$-cycles, and a $\neg\bar{c}$-permutation iff all its cycles are $\neg\bar{c}$-cycles.

Write $\pi = \pi_{\bar{c}} \circ \pi_{\neg\bar{c}}$ where $\pi_{\bar{c}}$ is the product of all $\bar{c}$-cycles of π, and $\pi_{\neg\bar{c}}$ is the product of all $\neg\bar{c}$-cycles of π. For a set $S \subseteq G(\mathbb{A})$, we write $\langle S\rangle$ for the subgroup generated by S, that is, the smallest subgroup of $G(\mathbb{A})$ containing S. In particular, if $S = \{\rho_1, \ldots, \rho_n\}$ is a finite set of permutations, we write $\langle \rho_1, \ldots, \rho_n\rangle$ for $\langle S\rangle$.

Complete author supporting proofs (publisher appendix A)

Lemma 2.3.4. Let $x \in \mathcal{X}$ and $\pi, \eta_1, \ldots, \eta_n$ be permutations written as compositions of disjoint cycles.

(1) If $\pi\cdot x = x$, η is a cycle of π, and $\mathrm{mv}(\eta)-\mathrm{supp}(x) \neq \varnothing$, then $\mathrm{mv}(\eta)\#\,\mathrm{supp}(x)$.
(2) If $\eta_i \cdot x = x$, for all $i = 1, \ldots, n$, and $\rho \in \langle \eta_1, \ldots, \eta_n\rangle$, then $\rho \cdot x = x$.

Proof. (1) Write $\pi = \mu_1 \circ \cdots \circ \mu_n$ with disjoint cycles $\mu_1, \ldots, \mu_n$, and let $\eta = \mu_j$ for some $1 \leq j \leq n$. Choose $a \in \mathrm{mv}(\eta) \setminus \mathrm{supp}(x)$. Suppose, by contradiction, there exists $b \in \mathrm{mv}(\eta) \cap \mathrm{supp}(x)$. Since the cycles are disjoint, $\mu_i(b) = b$ for all $i \neq j$. Hence $\pi(b) = \eta(b)$. Since $b \in \mathrm{supp}(x)$ and $\pi \cdot x = x$, it follows that $\pi(b) \in \mathrm{supp}(\pi \cdot x) = \mathrm{supp}(x)$, and therefore $\eta(b) \in \mathrm{supp}(x)$. Because $a, b \in \mathrm{mv}(\eta)$, there exists $l > 0$ such that $\eta^l(b) = a$. Applying π repeatedly to $\eta(b) \in \mathrm{supp}(x)$ and using that $\pi\cdot x = x$ yields $\eta^l(b) \in \mathrm{supp}(x)$, hence $a \in \mathrm{supp}(x)$, contradicting the choice of a.

(2) Assume that $\eta_i \cdot x = x$ for every $i = 1, \ldots, n$, and let $\rho \in \langle \eta_1, \ldots, \eta_n\rangle$. By definition of the subgroup generated by $\eta_1, \ldots, \eta_n$, the permutation ρ can be written as a finite composition $\rho = \zeta_1 \circ \cdots \circ \zeta_k$, where each ζ_r belongs to $\{\eta_1^{\pm 1}, \ldots, \eta_n^{\pm 1}\}$. We first note that each inverse η_i^{-1} also fixes x. Indeed, from $\eta_i \cdot x = x$, applying η_i^{-1} to both sides gives $\eta_i^{-1} \cdot (\eta_i \cdot x) = \eta_i^{-1} \cdot x$, and therefore $x = \eta_i^{-1} \cdot x$. So every ζ_r in the above decomposition fixes x. We now prove that ρ fixes x by repeated application of these equalities:

$$\rho \cdot x = (\zeta_1 \circ \cdots \circ \zeta_k) \cdot x = \zeta_1 \cdot (\zeta_2 \cdot (\cdots (\zeta_k \cdot x) \cdots)).$$

Since $\zeta_k \cdot x = x$, the innermost term simplifies to x; then $\zeta_{k-1} \cdot x = x$, and continuing in this way we obtain $\rho \cdot x = x$.

□

A key property of the N-quantifier is the Some/Any principle [16, Prop. 4.10]:
Proposition A.1 (Some/Any Principle). The following are equivalent:

$$\mathsf{N}\, c.\phi(c, \bar{x}) \iff \forall c.(c\#\bar{x} \Rightarrow \phi(c, \bar{x})) \iff \exists c.(c\#\bar{x} \wedge \phi(c, \bar{x})).$$

For multiple atoms, nested N-quantifiers introduce distinctness conditions. We avoid writing these explicitly by fixing the following convention.
Notation A.2. Let $\bar{c} = c_1, \ldots, c_n$ be a list of pairwise distinct atoms. We abbreviate $\forall c_1 \cdots \forall c_n.(\bigwedge_{i<j} c_i \neq c_j \Rightarrow \phi)$ as $\forall \bar{c}.\phi$, and similarly for $\exists \bar{c}.\phi$.

Lemma A.3 (Generalised Some/Any Principle). Let $\bar{c}$ be a list of atoms. Then:

$$\mathsf{N}\,\bar{c}.\phi(\bar{c},\bar{x}) \iff \forall\bar{c}.(\bar{c}\#\bar{x} \Rightarrow \phi(\bar{c},\bar{x})) \iff \exists\bar{c}.(\bar{c}\#\bar{x} \wedge \phi(\bar{c},\bar{x})).$$

Proof. By induction on the length of $\bar{c}$. The base case is (4). For the inductive step, write $\bar{c}' = \bar{c}, c_{n+1}$ and define $\psi(\bar{c},\bar{x}) := \mathsf{N}\,c_{n+1}.\phi(\bar{c},c_{n+1},\bar{x})$. Then $\mathsf{N}\,\bar{c}'.\phi \equiv \mathsf{N}\,\bar{c}.\psi$ and by the induction hypothesis and (4) this is equivalent to $\forall\bar{c}\ \forall c_{n+1}.(\bar{c}\#\bar{x} \wedge c_{n+1}\#\bar{x}, \bar{c} \Rightarrow \phi)$. By Notation A.2, this is exactly $\forall\bar{c}'.(\bar{c}'\#\bar{x} \Rightarrow \phi)$, as required. The existential case is analogous. □

Theorem 2.3.5 (Generalised Fix-Point). Let $x \in \mathcal{X}$ and let π be a permutation.

(1) If π is a $\bar{c}$-permutation and $A = \mathrm{mv}(\pi) \setminus \{\bar{c}\}$, then $\mathsf{N}\,\bar{c}.\pi \cdot x = x$ iff $A\#x$.
(2) $\mathsf{N}\,\bar{c}.\pi \cdot x = x$ iff $(\mathsf{N}\,\bar{c}.\pi_{\bar{c}} \cdot x = x) \wedge (\pi_{\neg\bar{c}} \cdot x = x)$. Equivalently, if $A = \mathrm{mv}(\pi_{\bar{c}}) \setminus \{\bar{c}\}$, then $\mathsf{N}\,\bar{c}.\pi \cdot x = x$ iff $A\#x \wedge (\pi_{\neg\bar{c}} \cdot x = x)$.

Proof. (1) ($\Rightarrow$) Assume $\mathsf{N}\,\bar{c}.\pi \cdot x = x$. By Lemma A.3, for some (equivalently, any) choice of $\bar{c}$ such that $\bar{c}\#x$, we have $\pi \cdot x = x$. Let η be a cycle of π. Since π is a $\bar{c}$-permutation, η contains some atom $a \in \bar{c}$. As $a\#x$, Lemma 1 yields $\mathrm{mv}(\eta)\#x$. Hence $\mathrm{mv}(\pi)\#x$, and therefore $A\#x$. ($\Leftarrow$) Assume $A\#x$. Let $\bar{c}$ be such that $\bar{c}\#x$. Since π is a $\bar{c}$-permutation, every cycle of π contains an atom from $\bar{c}$ and all remaining atoms lie in A. Thus every cycle is disjoint from $\mathrm{supp}(x)$, so $\pi \cdot x = x$. By Lemma A.3, it follows that $\mathsf{N}\,\bar{c}.\pi \cdot x = x$.

(2) If $\pi_{\bar{c}} = \mathsf{id}$ or $\pi_{\neg\bar{c}} = \mathsf{id}$, the result is immediate. Assume both are non-identity. We first show that $\mathsf{N}\,\bar{c}.\pi \cdot x = x$ iff $\mathsf{N}\,\bar{c}.(\pi_{\bar{c}} \cdot x = x \wedge \pi_{\neg\bar{c}} \cdot x = x)$. By Lemma A.3, both sides are equivalent to universal statements over $\bar{c}$ with the freshness condition $\bar{c}\#x, \mathrm{mv}(\pi) \setminus \{\bar{c}\}$. Hence it suffices to show that, for such $\bar{c}$, one has $\pi \cdot x = x$ iff both $\pi_{\bar{c}} \cdot x = x$ and $\pi_{\neg\bar{c}} \cdot x = x$. Assume $\pi \cdot x = x$. Let η be a cycle of $\pi_{\bar{c}}$. Since $\pi_{\bar{c}}$ is a $\bar{c}$-permutation, η contains some atom $a \in \bar{c}$. As $a\#x$, Lemma 1 yields $\mathrm{mv}(\eta)\#x$. Since this holds for every cycle η of $\pi_{\bar{c}}$, and the cycles are disjoint, we obtain $\mathrm{mv}(\pi_{\bar{c}})\#x$, and therefore $\pi_{\bar{c}} \cdot x = x$ (equivalently, $\pi_{\bar{c}}^{-1} \cdot x = x$). From $\pi = \pi_{\bar{c}} \circ \pi_{\neg\bar{c}}$ and $\pi \cdot x = x$, we have $\pi_{\bar{c}} \cdot (\pi_{\neg\bar{c}} \cdot x) = x$. Applying $\pi_{\bar{c}}^{-1}$ to both sides and using that $\pi_{\bar{c}}^{-1} \cdot x = x$, it follows that $\pi_{\neg\bar{c}} \cdot x = x$. Conversely, if both $\pi_{\bar{c}}$ and $\pi_{\neg\bar{c}}$ fix x, then $\pi \cdot x = (\pi_{\bar{c}} \circ \pi_{\neg\bar{c}}) \cdot x = \pi_{\bar{c}} \cdot (\pi_{\neg\bar{c}} \cdot x) = x$. From the fact that N distributes over conjunction, we obtain $\mathsf{N}\,\bar{c}.(\pi_{\bar{c}} \cdot x = x \wedge \pi_{\neg\bar{c}} \cdot x = x)$ is equivalent to $\mathsf{N}\,\bar{c}.(\pi_{\bar{c}} \cdot x = x) \wedge \mathsf{N}\,\bar{c}.(\pi_{\neg\bar{c}} \cdot x = x)$. Since $\pi_{\neg\bar{c}}$ does not involve atoms from $\bar{c}$, the formula $\pi_{\neg\bar{c}} \cdot x = x$ is independent of $\bar{c}$, and hence, $\mathsf{N}\,\bar{c}.(\pi_{\neg\bar{c}} \cdot x = x)$ is equivalent to $\pi_{\neg\bar{c}} \cdot x = x$. Therefore, $\mathsf{N}\,\bar{c}.\pi \cdot x = x$ iff $\mathsf{N}\,\bar{c}.(\pi_{\bar{c}} \cdot x = x) \wedge (\pi_{\neg\bar{c}} \cdot x = x)$, and the final equivalence follows from item (1), applied to $\pi_{\bar{c}}$ and B.

□

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